

Terna Engineering College

Computer Engineering Department

Program: Sem VI

Course: Cloud Computing Lab (CSL603)

Faculty: Preeti Patil

Experiment No.3

A.1 Aim:

To study and Implement Infrastructure as a Service using AWS.

A.2 Prerequisite:

Understanding of Virtualization, Basics of Networking, Basics of security and privacy.

A.3 Objective:

To demonstrate the steps to create and run virtual machines inside Public cloud platform.

A.3 Outcome: (L02)

After successful completion of this experiment students will be able to demonstrate infrastructure as a service by launching EC2 instance.

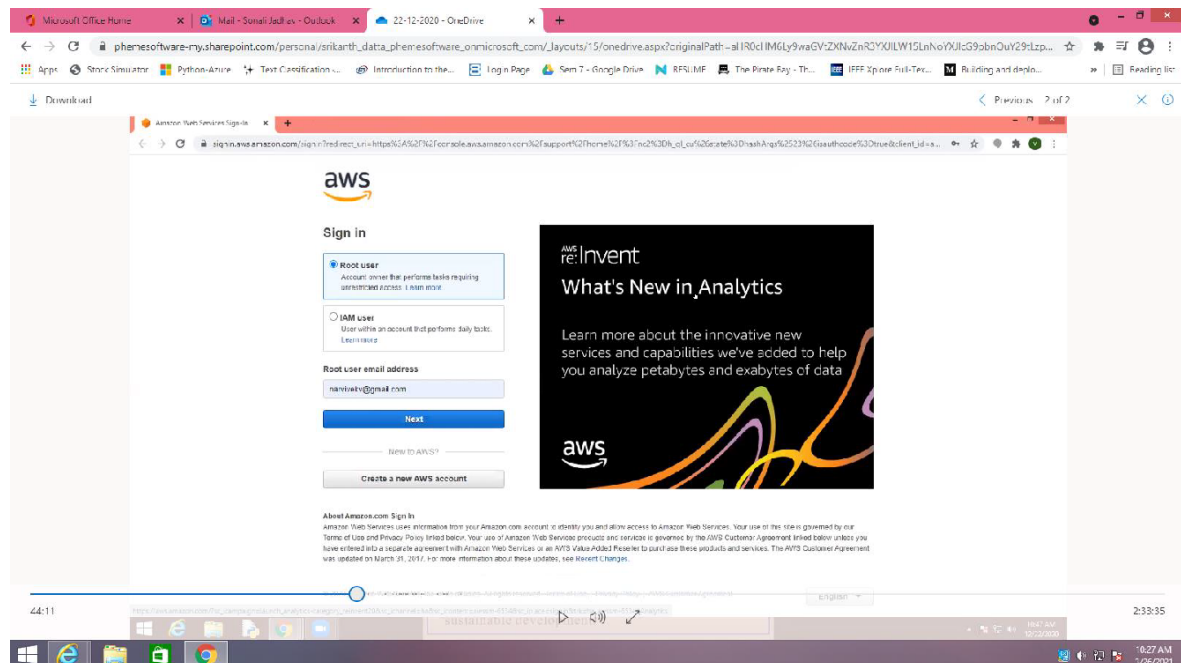
A.4 Theory:

Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides resizable compute capacity in the cloud. It is designed to make web-scale computing easier for developers and system administrators. Amazon EC2's simple web service interface allows obtaining and configuring capacity with minimal friction. It provides with complete control of computing resources and run on Amazon's proven computing environment. Amazon EC2 reduces the time required to obtain and boot new server instances to minutes, allowing you to quickly scale capacity, both up and down, as computing requirements change. Amazon EC2 changes the economics of computing by allowing you to pay only for capacity that actually use. Amazon EC2 provides developers and system administrators the tools to build failure resilient applications and isolate themselves from common failure scenarios. Auto Scaling allows scaling Amazon EC2 capacity up or down automatically according to conditions you define. With Auto Scaling, you can ensure that the number of Amazon EC2 instances you're using increases seamlessly during demand spikes to maintain performance, and decreases automatically during demand lulls to minimize costs. Elastic Load Balancing automatically distributes incoming application traffic across multiple Amazon

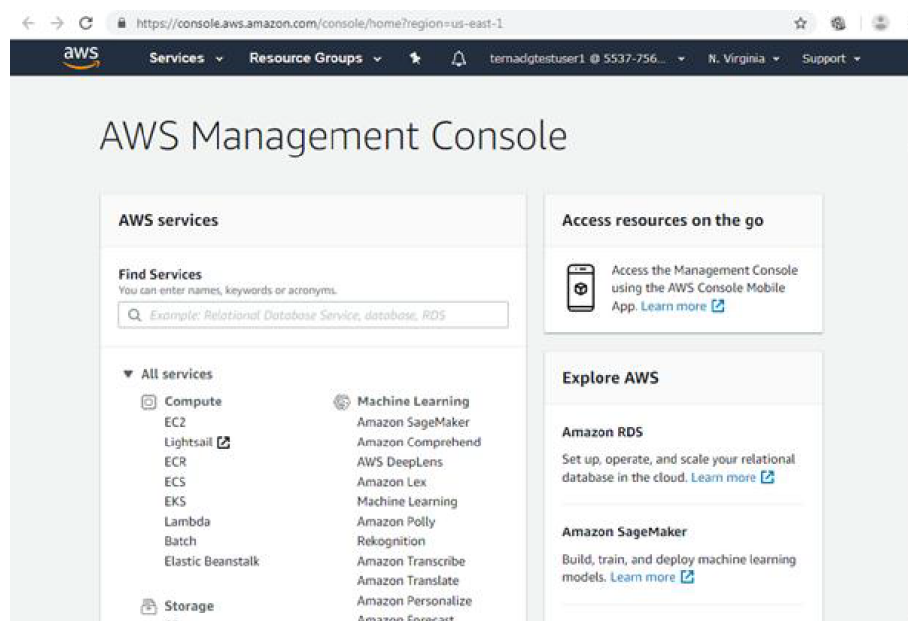
EC2 instances. It enables you to achieve even greater fault tolerance in your applications, seamlessly providing the amount of load balancing capacity needed in response to incoming application traffic.

Following are the steps to launch EC2 instances.

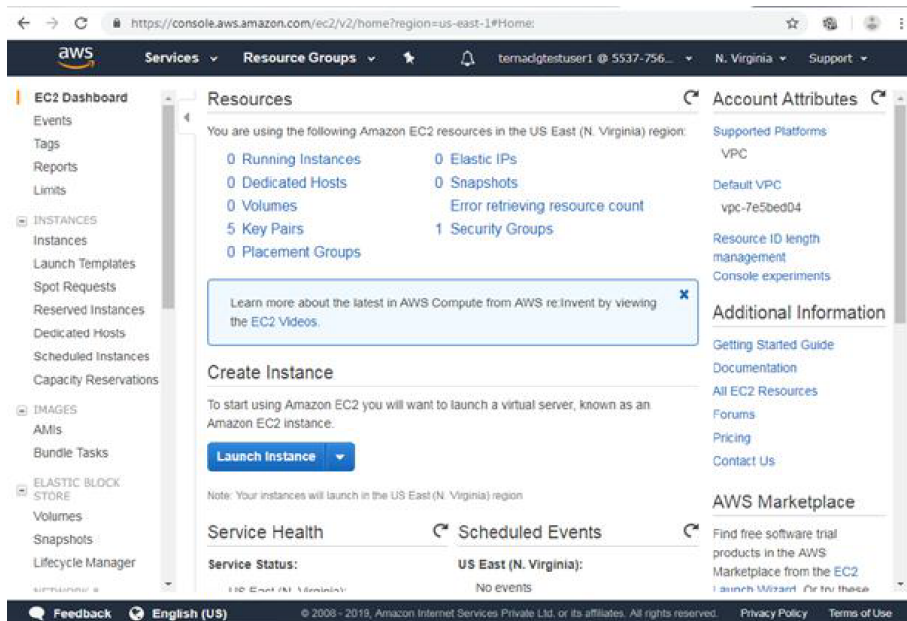
AWS account creation steps,



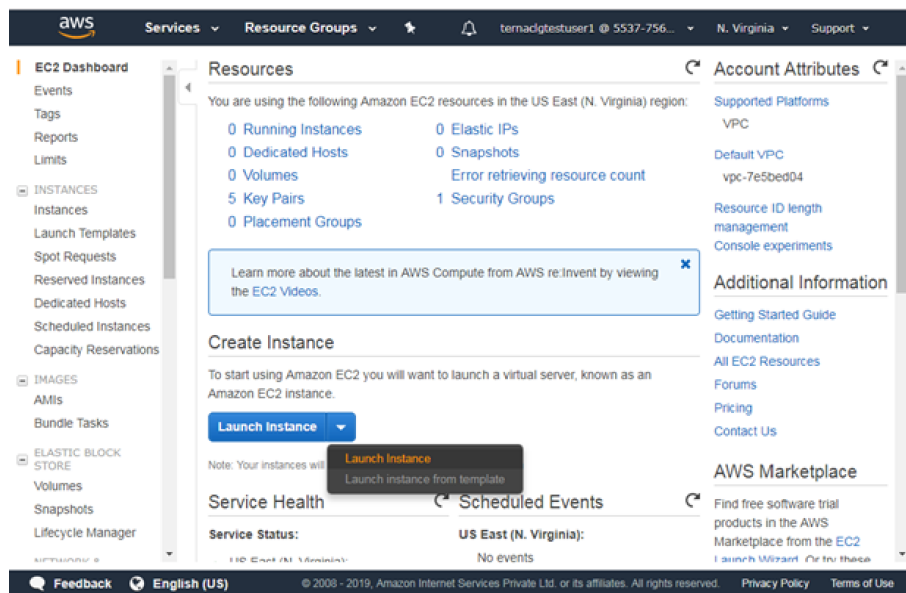
Step 1: Login



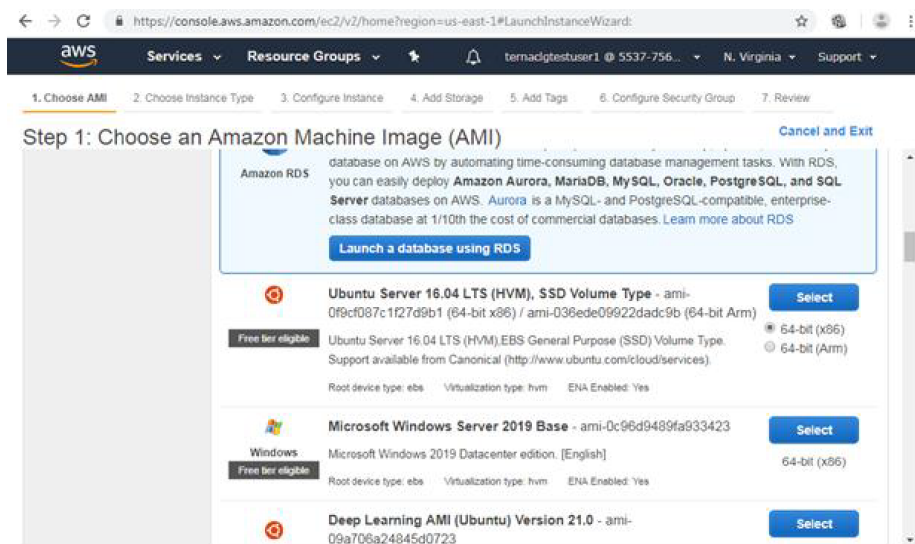
Step 2: Open AWS services



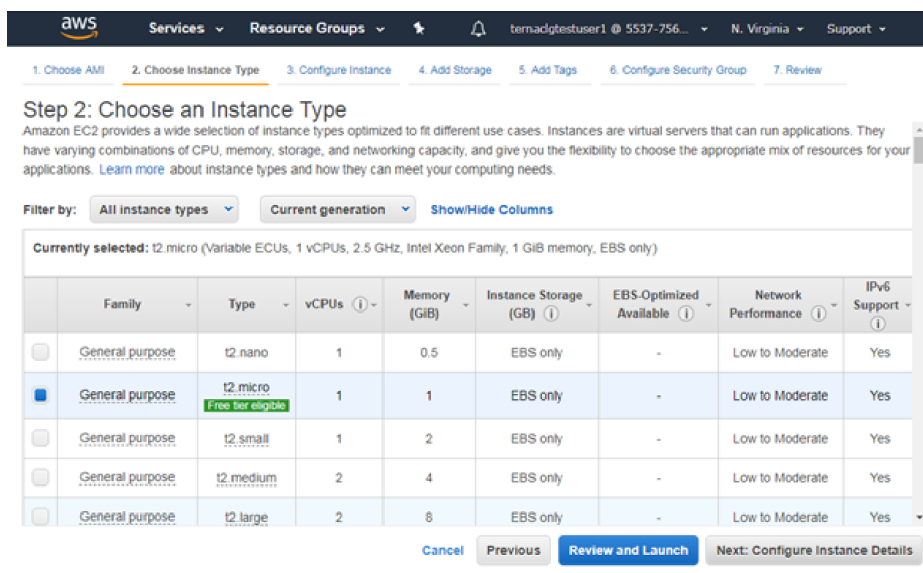
Step 3: Open EC2



Step 4: Select Launch Instance



Step 5: Select Amazon Linux



Step 6: Select Instance Type

aws

Services

Resource Groups

ternadgtester1 @ 5537-756...

N. Virginia

Support

1. Choose AMI

2. Choose Instance Type

3. Configure Instance

4. Add Storage

5. Add Tags

6. Configure Security Group

7. Review

Step 3: Configure Instance Details

lower pricing, assign an access management role to the instance, and more.

Number of instances

1

Launch into Auto Scaling Group

Purchasing option

☐ Request Spot instances

Network

vpc-7e5bed04 (default)

Create new VPC

Subnet

No preference (default subnet in any Availability Zone)

Create new subnet

Auto-assign Public IP

Use subnet setting (Enable)

Placement group

☐ Add instance to placement group

Capacity Reservation

Open

Create new Capacity Reservation

Domain join directory

No directory

Create new directory

Cancel

Previous

Review and Launch

Next: Add Storage

Step 7: Select Next: Add storage

aws

Services

Resource Groups

ternadgtester1 @ 5537-756...

N. Virginia

Support

1. Choose AMI

2. Choose Instance Type

3. Configure Instance

4. Add Storage

5. Add Tags

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7. Review

Step 4: Add Storage

Your instance will be launched with the following storage device settings. You can attach additional EBS volumes and instance store volumes to your instance, or edit the settings of the root volume. You can also attach additional EBS volumes after launching an instance, but not instance store volumes. [Learn more](#) about storage options in Amazon EC2.

Volume Type	Device	Snapshot	Size (GiB)	Volume Type	IOPS	Throughput (MB/s)	Delete on Termination	Encrypted
Root	/dev/sda1	snap-00b261520710021d8	30	General Purpose	100 / 3000	N/A	☒	Not Encrypted

Add New Volume

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage. [Learn more](#) about free usage tier eligibility and usage restrictions.

Cancel

Previous

Review and Launch

Next: Add Tags

Feedback

English (US)

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Step 8: Select Next: Add Tags

aws

Services

Resource Groups

ternadgtester1 @ 5537-756...

N. Virginia

Support

1. Choose AMI

2. Choose Instance Type

3. Configure Instance

4. Add Storage

5. Add Tags

6. Configure Security Group

7. Review

Step 5: Add Tags

A tag consists of a case-sensitive key-value pair. For example, you could define a tag with key = Name and value = Webserver.

A copy of a tag can be applied to volumes, instances or both.

Tags will be applied to all instances and volumes. [Learn more](#) about tagging your Amazon EC2 resources.

Key	Value	Instances	Volumes
Name	WindowsServer	☒	☒

Add another tag (Up to 50 tags maximum)

Cancel

Previous

Review and Launch

Next: Configure Security Group

Step 9: Select Next: Configure Security Group

aws

Services

Resource Groups

ternadgtester1 @ 5537-756...

N. Virginia

Support

1. Choose AMI

2. Choose Instance Type

3. Configure Instance

4. Add Storage

5. Add Tags

6. Configure Security Group

7. Review

Step 6: Configure Security Group

A security group is a set of firewall rules that control the traffic for your instance. On this page, you can add rules to allow specific traffic to reach your instance. For example, if you want to set up a web server and allow Internet traffic to reach your instance, add rules that allow unrestricted access to the HTTP and HTTPS ports. You can create a new security group or select from an existing one below. [Learn more](#) about Amazon EC2 security groups.

Assign a security group:

☒ Create a new security group

☐ Select an existing security group

Security group name: launch-wizard-2

Description: launch-wizard-2 created 2019-02-01T20:48:28.617+05:30

Type	Protocol	Port Range	Source	Description
RDP	TCP	3389	Custom 0.0.0.0/0	e.g. SSH for Admin Desktop

Add Rule

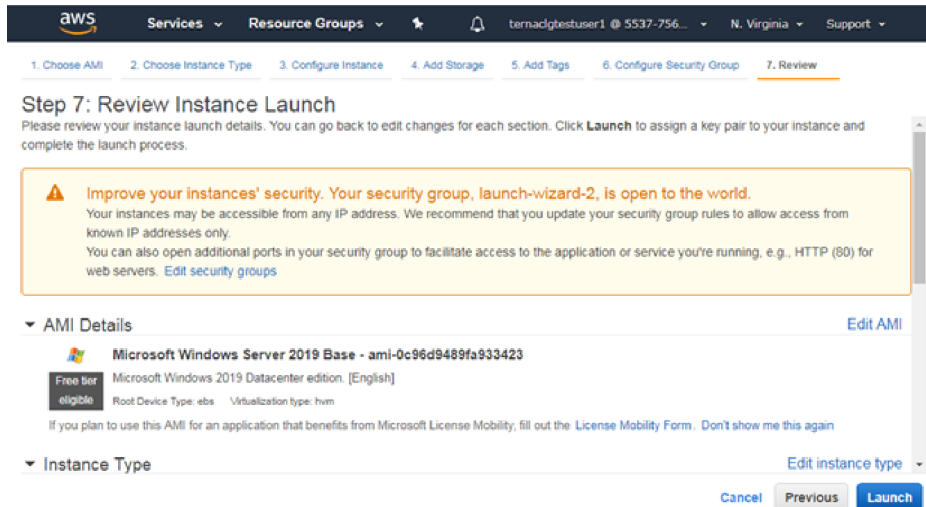
Warning

Cancel

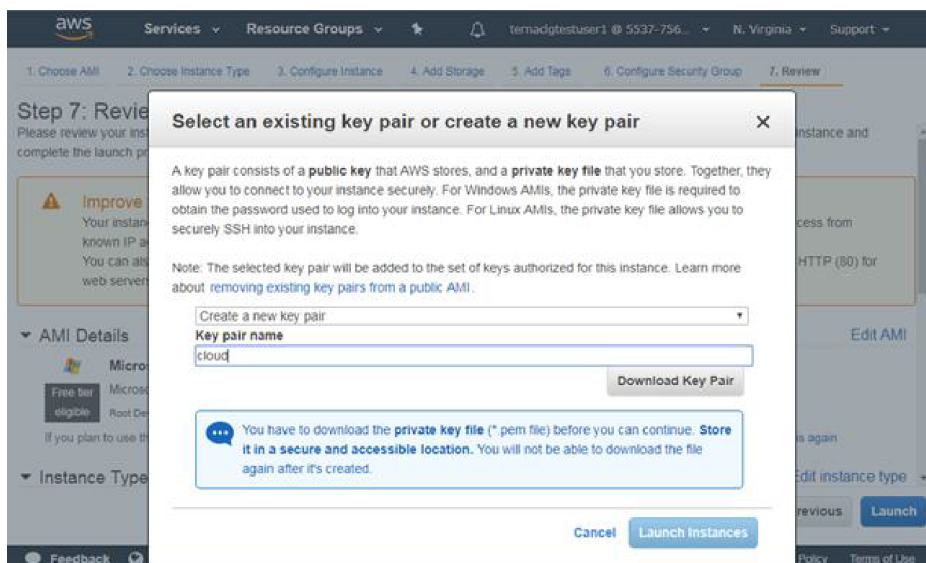
Previous

Review and Launch

Step 10: Select Review and launch



Step 11: Select launch



Step 12: Select an existing key pair and then click Launch

aws

Services

Resource Groups

ternadgtestuser1 @ 5537-756...

N. Virginia

Support

Launch Status

✓

Your instances are now launching

The following instance launches have been initiated: [i-08fba093071089f1d](#) [View launch log](#)

!

Get notified of estimated charges

Create [billing alerts](#) to get an email notification when estimated charges on your AWS bill exceed an amount you define (for example, if you exceed the free usage tier).

How to connect to your instances

Your instances are launching, and it may take a few minutes until they are in the **running** state, when they will be ready for you to use. Usage hours on your new instances will start immediately and continue to accrue until you stop or terminate your instances.

Click **View Instances** to monitor your instances' status. Once your instances are in the **running** state, you can **connect** to them from the Instances screen. [Find out](#) how to connect to your instances.

▼ Here are some helpful resources to get you started

[Amazon EC2: User Guide](#)

aws

Services

Resource Groups

ternadgtestuser1 @ 5537-756...

N. Virginia

Support

EC2 Dashboard

Events

Tags

Reports

Limits

INSTANCES

Instances

Launch Templates

Spot Requests

Reserved Instances

Dedicated Hosts

Scheduled Instances

Capacity Reservations

IMAGES

AMIs

Bundle Tasks

ELASTIC BLOCK STORE

Launch Instance

Connect

Actions

Filter by tags and attributes or search by keyword

< 1 to 2 of 2 >

	Name	Instance ID	Instance Type	Availability Zone	Instance State	Status Checks
☐	WindowsInst...	i-01d5635a2c710ec34	t2.micro	us-east-1a	terminated	
☒	WindowsSer...	i-08fba093071089f1d	t2.micro	us-east-1a	running	Initializing

Instance: **i-08fba093071089f1d** (Windows Server)

Public DNS: **ec2-52-23-167-230.compute-1.amazonaws.com**

Description

Status Checks

Monitoring

Tags

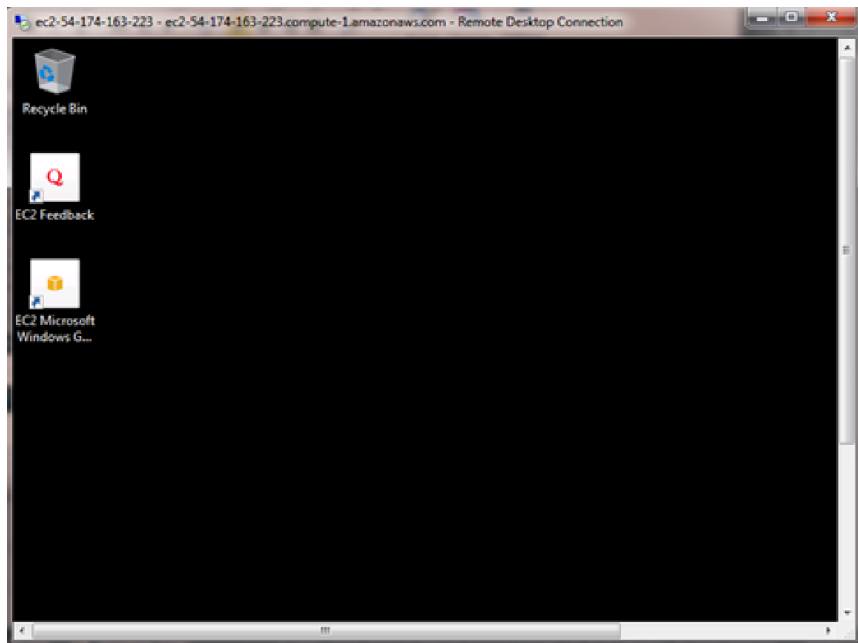
Instance ID

i-08fba093071089f1d

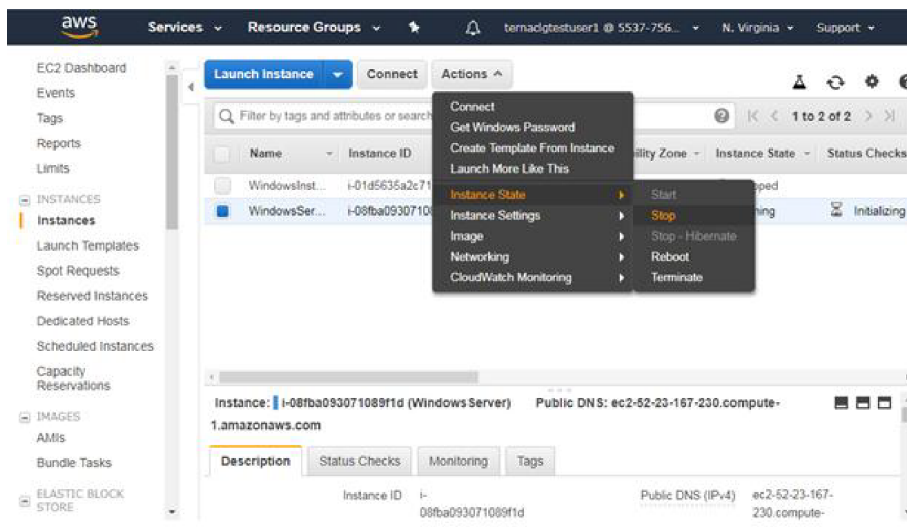
Public DNS (IPv4)

ec2-52-23-167-230.compute-

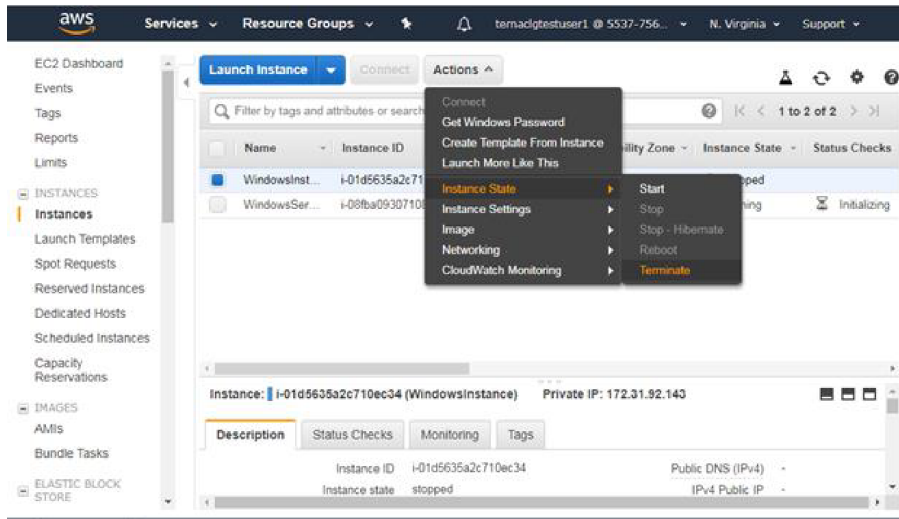
Step 13: Instance State then click Connect Instance



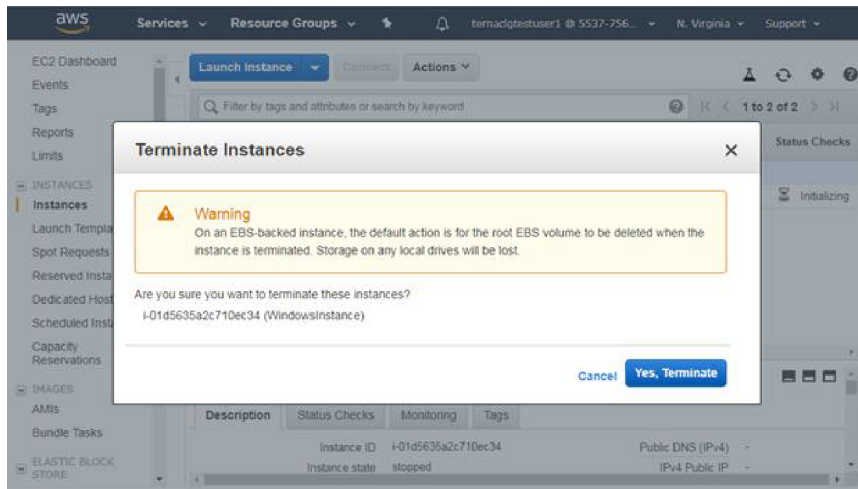
Step 14: After connecting instance



Step 15: How to stop instance



Step 16: How to terminate an instance



PART B

(PART B: TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the ERP or emailed to the concerned lab in charge faculties at the end of the practical in case there is no ERP access available)

Roll No.: C-56	Name: Quazi Md Moinuddin
Class : TE Comps C	Batch : C-3
Date of Experiment: 31.01.25	Date of Submission:5.02.25
Grade :	

B.1 Question of Curiosity:

Q1: What is Hypervisor? List out various hypervisors which are used to create Virtual machines

A **hypervisor** (also called a Virtual Machine Monitor, VMM) is software, firmware, or hardware that creates and manages **virtual machines (VMs)**. It allows multiple operating systems to share a single physical host by abstracting hardware resources and allocating them to different virtual instances.

Hypervisors are categorized into **two types**:

1. **Type 1 Hypervisor (Bare Metal)**: Runs directly on hardware, providing high performance and security.
 - Example: VMware ESXi, Microsoft Hyper-V, Xen, KVM
2. **Type 2 Hypervisor (Hosted)**: Runs on an existing operating system and uses it as a host to manage VMs.
 - Example: Oracle VirtualBox, VMware Workstation, QEMU

Popular Hypervisors Used for Creating Virtual Machines

1. **VMware ESXi** – Enterprise-grade Type 1 hypervisor for data centers.
2. **Microsoft Hyper-V** – Type 1 hypervisor built into Windows Server.
3. **KVM (Kernel-based Virtual Machine)** – Open-source Linux-based hypervisor (Type 1).
4. **Xen** – Open-source Type 1 hypervisor used in cloud computing.
5. **Oracle VirtualBox** – Type 2 hypervisor, great for testing and development.
6. **VMware Workstation & Player** – Type 2 hypervisor for personal and development use.
7. **QEMU** – Open-source emulator and hypervisor, widely used for system emulation.

Q2: Compare and Contrast Hardware and Software Virtualization.

Feature	Hardware Virtualization	Software Virtualization
Definition	Uses a hypervisor to create and manage virtual machines (VMs) directly on physical hardware.	Uses software to emulate hardware, allowing multiple OS instances to run on a single system.
Performance	High, as VMs run directly on the hardware.	Slower, as software emulation introduces overhead.
Hypervisor Type	Type 1 (Bare Metal)	Type 2 (Hosted)
Hardware Dependency	Requires CPU support for virtualization (e.g., Intel VT-x, AMD-V).	Can run on standard hardware without special virtualization features.
Isolation	Stronger isolation since VMs interact directly with hardware.	Weaker isolation due to reliance on the host OS.
Examples	VMware ESXi, Microsoft Hyper-V, KVM, Xen	VirtualBox, VMware Workstation, QEMU (with full emulation)

Use Case Data centers, cloud computing, Software development, testing, enterprise applications. legacy application support.

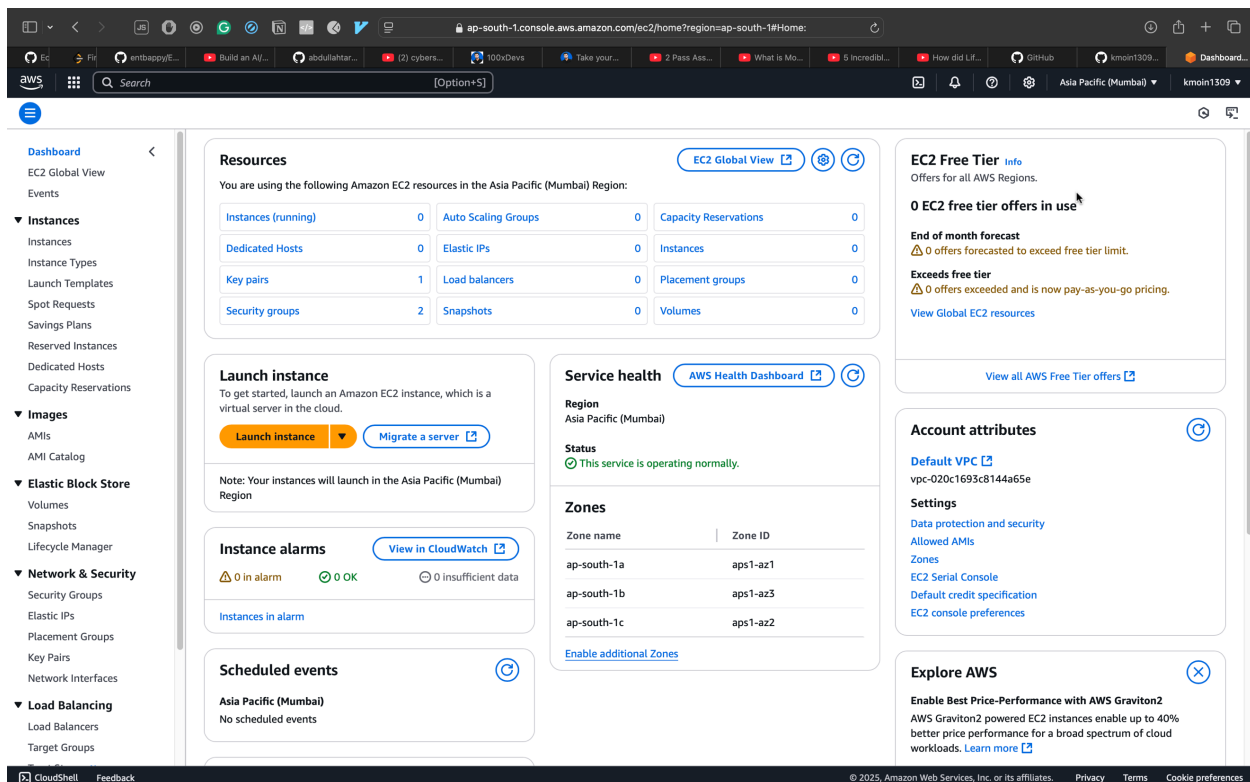
Flexibility Less flexible, as it depends on hardware support. More flexible, as it can run on a variety of systems.

Q3: Write down steps to install VM ware workstation (with appropriate screen shot).

Q4: Write down steps to create virtual machines (with appropriate screen shot).

Step 1: Log in to AWS Management Console

1. Go to [AWS Management Console](#).
2. Sign in with your **AWS credentials**.
3. Navigate to **EC2 Dashboard**.



Step 2: Launch a New EC2 Instance

1. Click on **Launch Instance**.
2. Enter an **Instance Name** (e.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Recents **Quick Start**

Amazon Linux **macOS** **Ubuntu** **Windows** **Red Hat** **SUSE Linux** **Debian**

Amazon Machine Image (AMI)

Amazon Linux 2023 AMI Free tier eligible

ami-05fa46471b02db0ce (64-bit (x86), uefi-preferred) / ami-0b0c4d81950666dc (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.6.20250128.0 x86_64 HVM kernel-6.1

Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.6.2...[read more](#)
ami-05fa46471b02db0ce

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

[Cancel](#) [Launch instance](#) [Preview code](#)

Architecture **Boot mode** **AMI ID** **Username** [Info](#)

CloudShell Feedback

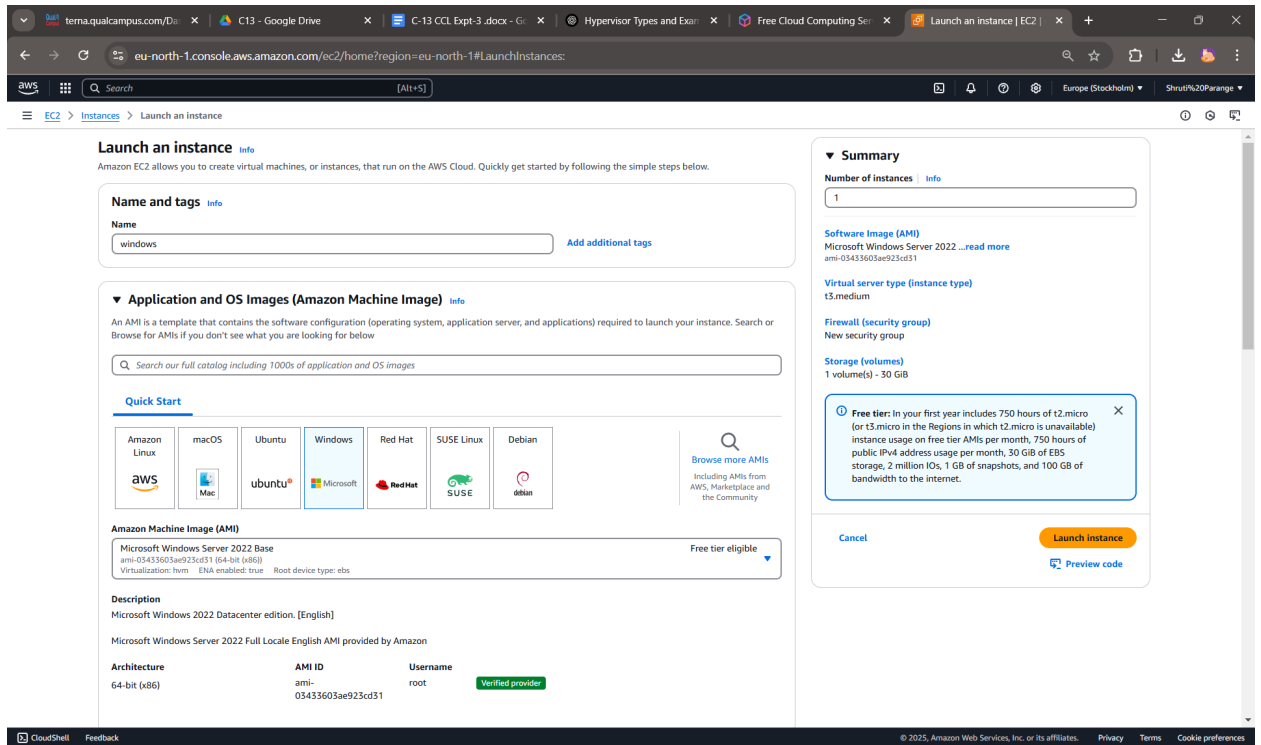
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g., "MyVM")

The screenshot displays the AWS Management Console interface for launching an EC2 instance. The 'Launch an instance' page is active, showing the 'Amazon Linux 2023 AMI' selected under the 'Quick Start' tab. The 'Instance type' is set to 't2.micro'. The 'Summary' panel on the right shows the configuration: 1 instance, Amazon Linux 2023 AMI, t2.micro instance type, new security group, and 1 volume (8 GiB). A 'Free tier' callout box indicates that the first year includes 750 hours of t2.micro usage. The 'Launch instance' button is visible at the bottom right.

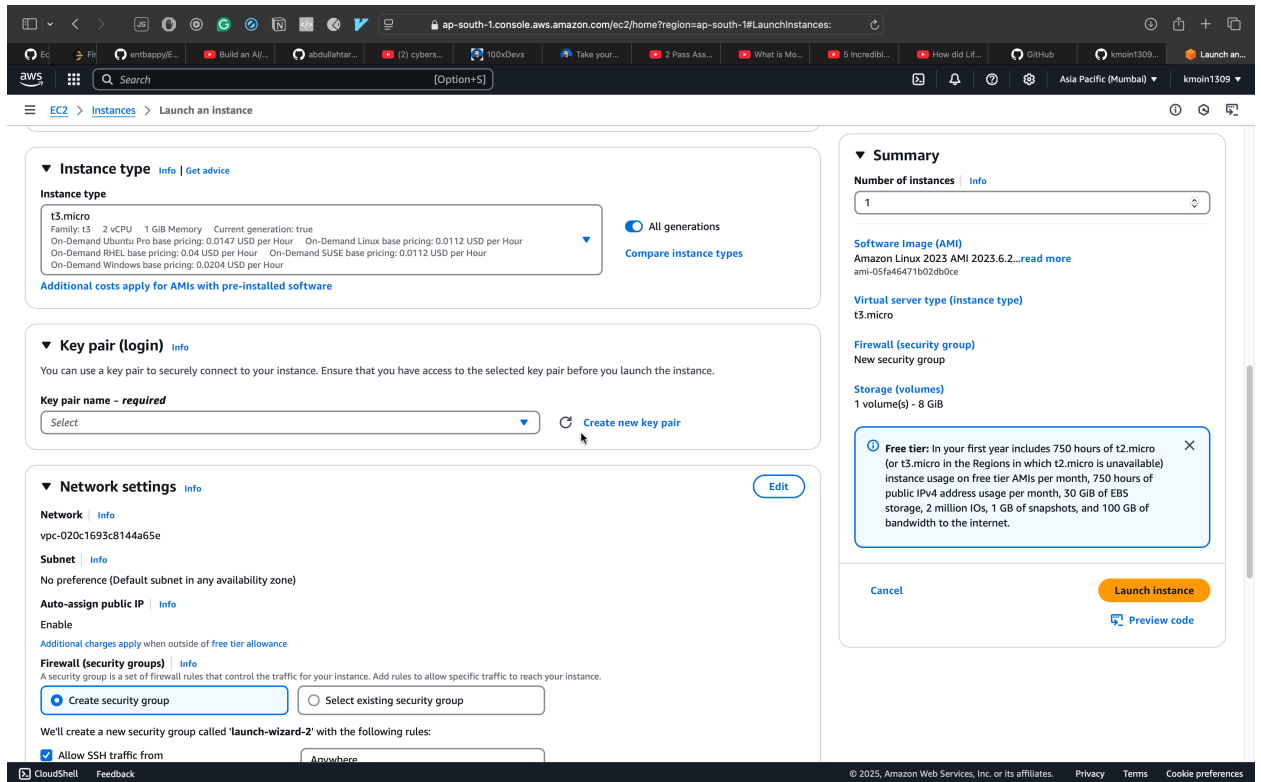
Step 3: Choose an Amazon Machine Image (AMI)

1. Select an **Operating System (AMI)** (e.g., **Amazon Linux, Ubuntu, Windows Server**).
2. Click **Select**.



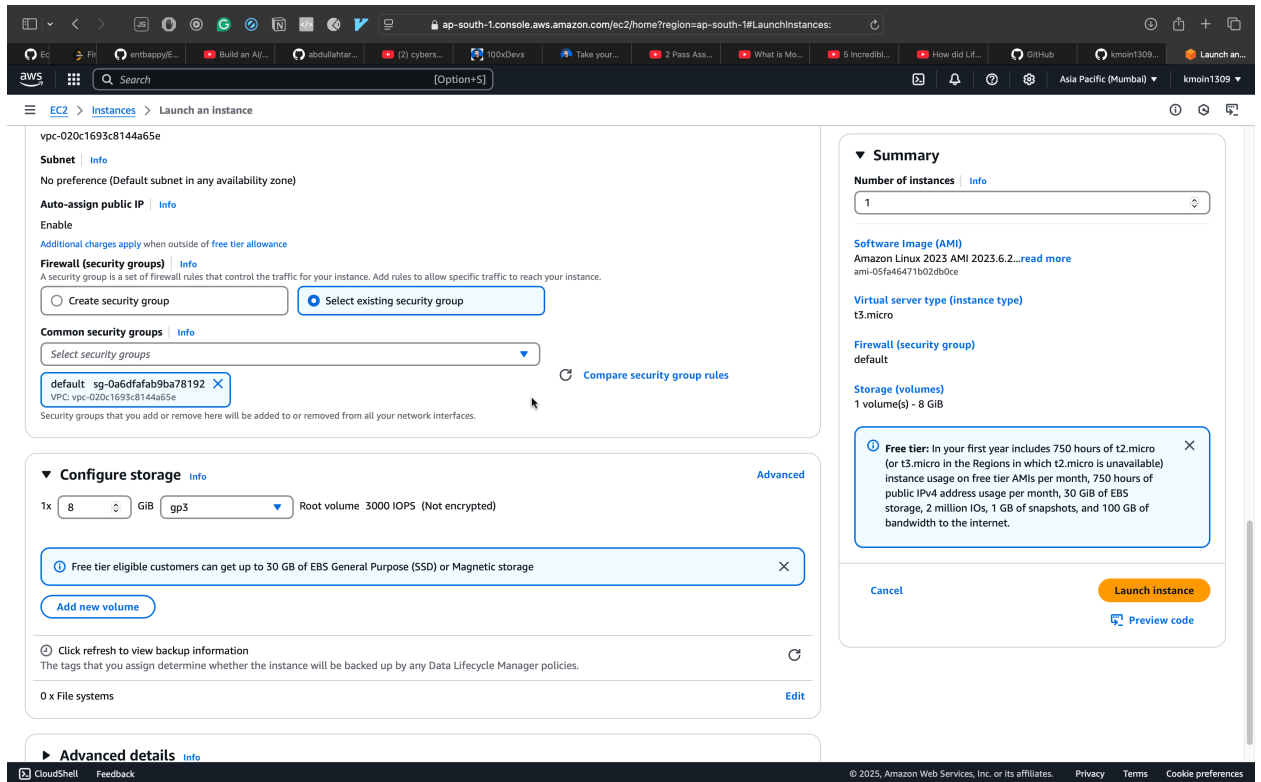
Step 4: Select an Instance Type

1. Choose a **VM configuration** (CPU, RAM).
2. Recommended: **t3.micro** (Free Tier Eligible).
3. Click **Next**.



Step 5: Configure Instance Details (Optional)

- Configure advanced settings like **IAM** roles, **Auto Scaling**, and **networking**.
- Click **Next**.



Step 6: Review and Launch

1. Review the instance details.
2. Click **Launch**.

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

EC2 > Instances > Launch an instance

Success

Successfully initiated launch of instance (i-0f553a45f612d4a4f)

Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing and free tier usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Learn more

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create a new RDS database

Learn more

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

Manage detailed monitoring

Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.

Manage detailed monitoring

Create Load Balancer

Create an application, network gateway or classic Elastic Load Balancer

Create Load Balancer

Create AWS budget

AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

Create AWS budget

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the instance.

Manage CloudWatch alarms

View all instances

CloudShell Feedback

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ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#instances:

Instances

Dashboard

EC2 Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Instances (1) info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Find Instance by attribute or tag (case-sensitive)

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IP
☐	MyVM	i-0f553a45f612d4a4f	Running	t3.micro	Initializing	View alarms +	ap-south-1a	ec2-106-153-38.ap-...	15.

Select an instance

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Q.5 What do you mean by system image?

A **system image** is a complete backup of an entire computer system, including the **operating system (OS), system settings, installed applications, and all user files**. It allows you to restore the system to a previous state in case of system failure, hardware crash, or software corruption.

Key Features of a System Image:

Full Backup – Contains everything on the system, including boot files.

Restoration – Can be used to restore the system to an exact previous state.

Bootable – Some system images can be booted directly to reinstall the OS.

File Formats – Common formats include **.iso**, **.img**, **.vhd**, **.vmdk**, and **.tib**.

Q.6 List out steps to create system image backup.

Method 1: Create a System Image Backup in Windows

(Using Built-in Windows Backup & Restore)

Step 1: Open Backup Settings

1. Open **Control Panel** → Go to **Backup and Restore (Windows 7)**.
2. Click **Create a system image** in the left panel.

Step 2: Choose a Backup Location

You will be asked **where to save the system image:**
External Hard Drive (Recommended)
Network Location
DVDs (not recommended due to size limitations)

Step 3: Select Drives to Backup

- By default, Windows selects the **system drive (C:)**.
- You can add additional drives if needed.

Step 4: Start the Backup Process

- Click **Next**, then **Start Backup**.
- Wait for the process to complete (it may take time depending on system size).

Step 5: Create a System Repair Disk (Optional but Recommended)

- Windows will ask if you want to create a **System Repair Disk**.
- This helps in restoring the image in case of failure.
- You can also create a **bootable USB** instead.

B.2 Conclusion:

Infrastructure as a Service (IaaS) using AWS provides scalable, on-demand computing resources such as virtual machines (EC2), storage (S3, EBS), and networking (VPC). By implementing IaaS, organizations can reduce hardware costs, enhance flexibility, and improve system reliability. AWS simplifies infrastructure management, enabling businesses to focus on innovation rather than maintenance.